Virtual Walk-In Queue Management System - Phase 1 Design

Project Idea Origin

The idea originated from observing long waiting times at a barber shop. Customers were forced to remain physically present to keep their place in line. The goal is to reduce forced waiting time by allowing customers to keep their queue position while spending their waiting time elsewhere.

Core Objective

Reduce forced waiting, not eliminate waiting. Customers can join a queue, leave the premises if desired, and return when their turn approaches.

Customer Flow

Arrive at shop → Seats available?
Yes → Direct haircut.
No → Scan QR → Enter Name + Phone Number → Join Queue → Receive Tracking Link → View Live Queue Position → Notification (#2 in queue) → Notification (#1 in queue) → Return to shop → Press 'I've Reached The Shop' → Haircut.

Shop Flow

Login → View Active Chairs → View Waiting Queue → Chair Becomes Free → Press Next Customer → Confirmation Popup → Assign Customer → Queue Updates → Notifications Sent.

Queue Rules

• One shared queue
• First Come First Serve (FCFS)
• Multiple chairs supported
• No customer account required
• QR code scanned at the shop
• Name + phone number required
• Customer can leave after joining queue

Notifications

• When customer becomes #2 in queue: 'Your turn is approaching.'
• When customer becomes #1 in queue: 'You are next. Please reach the shop.'

Customer Statuses

WAITING, NOTIFIED, ARRIVED, SERVING, COMPLETED, SKIPPED

Data Retention Policy

- Active queue data is maintained live.
- Historical records are retained for 30 days.
- Automatic deletion after 30 days.

Future Phases

Database schema, API design, frontend screens, backend implementation, Socket.IO integration, notifications, deployment, and presentation materials.